



KERTORY
JEPa RESEARCH DIVISION

COMPLETE RESEARCH JOURNEY
INDEPENDENT RESEARCH PROJECT

KERTORY-JEPA

V0.1 → V0.2 → V0.3.1

Complete Research Journey

Predictive Latent World Model —
Physics Generalization Series

VERSIONS

3 (V0.1 → V0.3.1)

VARIABLES RANDOMIZED

0 → 1 → 4

EPISODES

300 → 500 → 1000

HEADLINE RESULT

Few-shot: -54.6% error



SECTION 01

Overview

This report documents the complete arc of the Kertory-JEPA physics-generalization series, from the first prototype (V0.1) through two subsequent iterations (V0.2, V0.3.1). Each version tested one specific hypothesis, built directly on what the previous version revealed, and is reported here with its actual measured results — including a clear negative result in V0.1 that directly motivated the design of V0.2 and V0.3.1.

THE CORE QUESTION ACROSS ALL THREE VERSIONS

Can a self-supervised, JEPA-style predictor learn reusable physics dynamics from observation, well enough to handle physical conditions it was never trained on?

How to read this report

- **One hypothesis per version**

Each version changes exactly one thing in response to what the previous version's evaluation showed, rather than changing many variables at once.

- **Negative results are kept in**

V0.1's failure is reported in full because it is what directly motivated V0.2's design — not smoothed over as a discarded draft.

- **Measured, not projected**

Every number in this report is a measured evaluation result. Interpretation is kept separate from and clearly labeled against the raw figures.

SCOPE

All results are from controlled physics simulations, not real-world data. Section 6 states plainly what this series has and has not demonstrated.



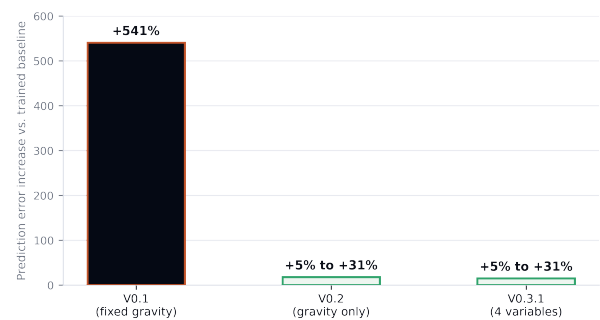
SECTION 02

Version summary at a glance

Three versions, each changing one thing at a time: what varies during training, how much data is used, and how the model is tested.

METRIC	V0.1	V0.2	V0.3.1
Variables randomized	0 (fixed g=1.0)	1 (gravity)	4 (grav, rest, radius, friction)
Episodes	300	500	1000
Parameters	~22.3M	~22.1M	~32–35M
Training val loss	0.1329	— (per-condition)	0.01871
In-range generalization	Failed (+541%)	Good (+5–31%)	Good (+5–31%)
Extrapolation (out-of-range)	Not tested	Weak (+85–135%)	Weak (+62–67%)
Few-shot adaptation test	Not tested	Not tested	Yes — 54.6% improvement

IN-RANGE ERROR INCREASE, BY VERSION



V0.2 introduced gravity randomization; V0.3.1 extended randomization to four physics variables while holding in-range error at a similarly low level.



SECTION 03 · V0.1

First prototype — setup

V0.1 established the baseline pipeline and tested the simplest possible version of the research question: a single, fixed physical condition, held constant across all training episodes.

ENVIRONMENT

A single ball falling under gravity and bouncing; gravity fixed at 1.0 for all training episodes.

ENCODER

Frozen Meta V-JEPA2 (facebook/vjepa2-vitl-fpc64-256).

PREDICTOR

22.28M-parameter JEPA-style transformer, trained from scratch.

DATASET

300 episodes, single gravity value, varied only in starting position.

Training result

TRAIN LOSS

0.1301

VALIDATION LOSS

0.1329

EPOCHS

30

Train and validation loss are close in value — 0.1301 vs. 0.1329 — showing no overfitting signal within the training distribution itself. The question this leaves open is what happens outside that distribution, tested next in Stage D.

WHAT V0.1 ESTABLISHED

Beyond the numeric result, V0.1 proved out the full pipeline used by every later version: raw physics simulation, a frozen V-JEPA2 encoder, a trained predictor, and a repeatable evaluation protocol. That pipeline carried forward unchanged into V0.2 and V0.3.1.



SECTION 03.2 · V0.1

Stage D — generalization test

The trained model was evaluated — frozen, no further training — on new episodes at gravity values it had never seen.

CONDITION	MSE	CHANGE VS. BASELINE
Gravity = 1.0 (trained)	0.139	baseline
Gravity = 0.5 (unseen)	0.894	+541%
Gravity = 2.0 (unseen)	0.733	+426%

REJECTED

Prediction error increased 4–5x when gravity changed. The model had learned statistics specific to gravity = 1.0, not a generalizable representation of gravity as a variable.

Root cause

The training data contained exactly one gravity value. The model was never shown variation, so it had no basis to represent gravity as something that could take different values.

This directly motivated V0.2: randomize gravity during training instead of holding it fixed.

WHY THIS RESULT IS REPORTED IN FULL

A weaker report might have quietly moved to a better configuration without showing this failure. It is included here because it is the reason V0.2 and V0.3.1 randomize their training variables at all — the negative result is part of the evidence, not a discarded draft.



SECTION 04 · V0.2

What changed from V0.1

V0.2 addressed the V0.1 root cause directly: gravity was randomized during training, and several supporting fixes were made to the loss and architecture.

- **Gravity randomized per episode**

Sampled from a range of 0.3 to 2.5, instead of fixed at 1.0.

- **Episodes increased**

From 300 to 500.

- **Temporal pooling improved**

Preserves 32 time-ordered positions; V0.1 used simple flat averaging, which lost motion order.

- **JEPA-style loss added**

LayerNorm-normalized Smooth-L1 loss plus a temporal-difference term, replacing plain MSE.

- **Mask-token bug fixed**

A single shared “mask token” for all future positions was replaced with per-position learned embeddings — the shared token gave the model no way to tell which future time-step it was predicting.

WHY THESE CHANGES TOGETHER

The core fix is randomizing gravity; the pooling, loss, and mask-token changes remove confounding issues that could otherwise have masked whether the core fix was working.



SECTION 04.2 · V0.2

Stage D — generalization test

Both within the training range (interpolation) and outside it (extrapolation) were tested.

CONDITION	LOSS	CHANGE VS. BASELINE
Gravity 1.0 (in-range, trained)	0.02718	baseline
Gravity 0.5 (in-range)	0.03572	+31%
Gravity 2.0 (in-range)	0.02848	+5%
Gravity 0.15 (OUT of range)	0.06386	+135%
Gravity 3.0 (OUT of range)	0.05019	+85%

CONFIRMED (IN-RANGE)

Within the training range, error increased only 5–31%, compared to 541% in V0.1 — a dramatic improvement. Outside the training range, error still increased substantially (85–135%), showing the model interpolates well but does not yet extrapolate well.

READING THE DIVIDING LINE

The blue rule in the table above marks the edge of the training range (0.3 to 2.5). Rows above it are interpolation; rows below it are extrapolation — a meaningfully harder test that V0.3.1 revisits directly.

What this leaves open

V0.2 shows the model can interpolate within a range it was trained across, but extrapolation beyond that range — and generalizing across multiple physics variables at once rather than gravity alone — remained untested. Both questions are taken up directly in V0.3.1, described next.



SECTION 05 · V0.3.1

Multi-variable physics + few-shot adaptation — what changed

V0.3.1 extends V0.2 along two axes at once: more physics variables randomized together, and a new test of whether the model can adapt quickly to an unfamiliar environment.

- **Three more variables randomized**

Restitution (bounciness, 0.4–0.95), ball radius (6–18 px), and friction (0.0–0.15) — up from 1 variable to 4.

- **Episodes increased**

From 500 to 1000, to cover the larger combined parameter space.

- **Model capacity increased**

From ~22M to ~32–35M parameters (predictor width 384→448).

- **Overfitting safeguards added**

Dropout (0.1) on attention and MLP layers, plus early stopping (patience = 5 epochs), to offset the larger model and more complex data.

- **Testing expanded to five stages (A–E)**

So that if generalization failed, the specific variable or combination responsible could be identified, rather than one opaque pass/fail result.

- **New Stage E: few-shot adaptation**

Tests whether the model, after seeing only a handful of examples from a new environment, adapts faster than starting from zero — without full retraining.

WHY STAGE E MATTERS

Stage E was designed to directly test the idea that a system should be able to adjust to an unfamiliar environment from a small number of observations, rather than needing to be retrained from scratch. Results are reported in Section 5.4.



SECTION 05.2–5.3 · V0.3.1

Training result & full Stage A–E results

EPOCHS COMPLETED

30

FINAL TRAIN LOSS

0.01922

FINAL VALIDATION LOSS

0.01871

Train and validation loss tracked closely throughout — no overfitting signal.

FULL STAGE A–E RESULTS

STAGE	GRAV	REST	RAD	FRIC	LOSS
A — baseline	1.40	0.68	12	0.075	0.02519
B — gravity	0.50	0.68	12	0.075	0.02579
B — gravity	2.00	0.68	12	0.075	0.02712
B — restitution	1.40	0.45	12	0.075	0.02226
B — restitution	1.40	0.90	12	0.075	0.03077
B — radius	1.40	0.68	7	0.075	0.02126
B — radius	1.40	0.68	17	0.075	0.02796
B — friction	1.40	0.68	12	0.020	0.02379
B — friction	1.40	0.68	12	0.130	0.02379
C — grav+rest	0.50	0.90	12	0.075	0.03388
C — grav+radius	2.00	0.68	16	0.075	0.02977
C — rest+radius	1.40	0.45	8	0.075	0.02217
C — grav+friction	1.40	0.68	12	0.120	0.02428
D — extrapolation	0.15	0.35	20	0.200	0.04196
E — zero-shot	0.15	0.35	20	0.200	0.04057
E — few-shot (8 ep)	0.15	0.35	20	0.200	0.01843

Columns: Grav = gravity, Rest = restitution, Rad = ball radius (px), Fric = friction. Orange rows mark the hardest out-of-range test; green marks the few-shot adaptation result detailed in Section 5.4.



SECTION 05.4 · V0.3.1

Interpretation, stage by stage

**Stage A — Baseline****Loss: 0.02519**

Center-of-range values — consistent with training performance.

**Stage B — One variable moved at a time****Loss: 0.021–0.031**

Losses stayed close to baseline for all four variables individually — each generalizes well on its own within the training range.

**Stage C — Two variables moved together****Loss: 0.022–0.034**

Losses remained in a similar band — the model handles pairs of variables changing jointly without a sharp breakdown.

**Stage D — All four variables pushed out of range****Loss: 0.04196**

The hardest test. A meaningful increase over baseline, but far smaller than V0.1's 541% failure — extrapolation is still harder than interpolation, as expected.

PATTERN ACROSS A–D

Error rises gradually from single-variable changes (B) through paired changes (C) to full out-of-range extrapolation (D) — a smooth degradation, not a sudden breakdown, and each stage isolates a specific cause rather than giving one opaque pass/fail result.

And then, Stage E

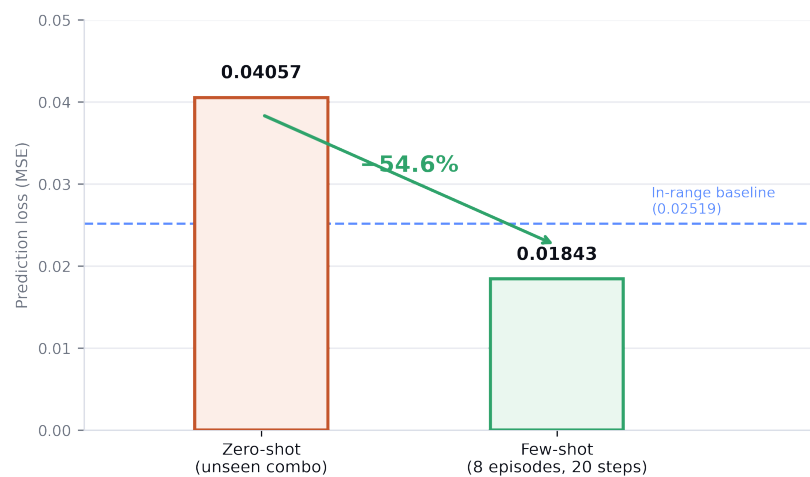
Stage E — few-shot adaptation — is treated separately on the next page. It is the most important result in this report: the first direct evidence that the model can improve its handling of an unfamiliar environment from a small number of observations, without full retraining.



THE MOST IMPORTANT RESULT IN THIS SERIES

Few-shot adaptation works

Zero-shot loss on the unseen variable combination was 0.04057. After showing the model just 8 new episodes and allowing a short, low-learning-rate adjustment (20 steps, not full retraining), loss dropped to 0.01843.



REDUCTION

-54.6%

NEW EXAMPLES SEEN

8

ADAPTATION STEPS

20

This is the first direct evidence in this project that the model can improve its handling of an unfamiliar environment from a small number of observations, without full retraining — the specific capability the project set out to test.



SECTION 06.1

Demonstrated

Across all three versions, this series has demonstrated the following, each backed by a measured result reported earlier in this document.

**A working, reproducible pipeline**

From raw physics simulation → frozen V-JEPA2 encoder → trained predictor → measured evaluation, used unchanged across all three versions.

**A diagnosed and directly addressed failure**

A specific, previously-identified failure — memorizing one fixed condition instead of generalizing — was diagnosed in V0.1 and directly addressed by changing the training data, not just the model.

**Generalization across up to four physics variables**

Within a trained range, covering gravity, restitution, radius, and friction simultaneously (V0.3.1, Stages A–C).

**A measurable few-shot adaptation effect**

A large error reduction (–54.6%) from a small number of new observations, without full retraining (V0.3.1, Stage E).

WHAT TIES THESE TOGETHER

Each demonstrated result traces back to a specific, named experiment earlier in this report — none are summary claims without a measured figure behind them.



SECTION 06.2

Not demonstrated

Stated plainly and without qualification, so that the results in this report are not read more broadly than they support.

**Strong extrapolation far outside the training distribution**

Error still rises substantially there (Stage D, +62–67%).

**Physics or causal understanding in any formal sense**

E.g., no explicit representation of force, mass, or energy conservation.

**Any real-world, non-simulated environment**

No real object, sensor, or robotics data was used anywhere in this series.

**General intelligence, or any comparison to language models**

No claim of, or comparison to, general intelligence or large language models is made or implied.

No result in this report should be read as implying any of the above.



SECTION 07

Next direction

The clearest open question this series leaves is extrapolation: performance degrades outside the trained range even though in-range generalization and few-shot adaptation both work well.

Candidate directions for a future version

1

Widen the training distribution further

Cover a broader range of each physics variable during training, directly attacking the extrapolation gap at its source.

2

Combine few-shot adaptation with out-of-range conditions

Test whether Stage E's adaptation mechanism can specifically close the extrapolation gap, rather than only helping within already-familiar territory.

3

Explore physics-structured architectures

E.g., energy-conservation-based models, as a complementary approach to a purely data-driven predictor.

Closing summary

Across V0.1, V0.2, and V0.3.1, prediction error under distribution shift fell from +541% to +62–67%, and a working few-shot adaptation mechanism was demonstrated for the first time in this project. Extrapolation beyond the trained range remains the open problem the next version should address.

Same pipeline, same evaluation protocol, three versions — each measured result building directly on what the last one showed.



KERTORY

JEPA RESEARCH DIVISION

Thank you for following this research journey.

V0.4 begins where V0.3.1 leaves off: extrapolation.

KERTORY-JEPA · V0.1 → V0.2 → V0.3.1 Research Journey

Author: Vikash Kumar · Independent Research Project